

Proof of Conjecture 00000006680

Two transport problems with the same optimal value and different dual structure

1 The conjecture and the reading

Definition: The optimal value and the dual structure are two layers. *Conjecture:* There exist two transport problems with the same optimal value but different dual structures, and the separation is realized by an explicit degenerate cost pair with the same value but different support.

We prove it for finite discrete (Kantorovich) transport. A transport problem is (μ, ν, c) with probability vectors $\mu \in \mathbb{R}^m$, $\nu \in \mathbb{R}^n$ and cost $c \in \mathbb{R}^{m \times n}$.

- A plan is $\pi \geq 0$ with row sums μ and column sums ν ; its cost is $\sum_{i,j} c_{ij} \pi_{ij}$; the *optimal value* is the infimum of this over plans (**optValue**).
- Dual: potentials (u, v) with $u_i + v_j \leq c_{ij}$; objective $\sum \mu_i u_i + \sum \nu_j v_j$; an optimal dual solution maximises it among feasible pairs (**IsDualOpt**). **StrongDuality:** some optimal dual solution attains the optimal value.
- *Dual structure* (layer two) is read as: the optimal dual solution is unique up to the gauge $(u, v) \mapsto (u + t, v - t)$ (**DualUniqueModGauge**); two problems have different dual structure if this holds for one and fails for the other.
- *Support* is read in two ways, both proved: (A) the family of supports of optimal plans (**optSuppFamily**); (B) the support $\{c_{ij} \neq 0\}$ of the cost matrix (**costSupp**).

The cost pair is degenerate (it has zero entries and admits several optimal plans).

2 The pair

Take $m = n = 2$, $\mu = \nu = (\frac{1}{2}, \frac{1}{2})$ and $C_{a,b} = \begin{pmatrix} 0 & 0 \\ a & b \end{pmatrix}$, $0 \leq a \leq b$. Plans are exactly $P_x = \begin{pmatrix} x & \frac{1}{2} - x \\ \frac{1}{2} - x & x \end{pmatrix}$, $0 \leq x \leq \frac{1}{2}$ (this is the one-parameter description of 2×2 doubly stochastic matrices used for Conjecture 00000006320, rescaled by $\frac{1}{2}$). Its cost is $a(\frac{1}{2} - x) + bx \geq a/2$, so the optimal value is $a/2$, attained at $x = 0$; P_x is optimal iff $(b - a)x = 0$.

Dual. For feasible (u, v) , $(u_0 + v_1) + (u_1 + v_0) \leq 0 + a$, so the objective is $\leq a/2$; $u = (0, a)$, $v = (0, 0)$ is feasible and attains it, so the dual value equals the primal value, and (u, v) is optimal iff it is feasible with $u_0 + v_1 = 0$, $u_1 + v_0 = a$.

Pair 1: $C_{0,0}$ (zero cost) and $C_{0,1}$. Both have value 0 and no duality gap.

- $C_{0,0}$: every P_x is optimal, in particular the full-support $P_{1/4}$. $C_{0,1}$: optimal plans need $x = 0$, so $P_{1/4}$ is not optimal, and no optimal plan has full support. Hence the families of optimal supports differ (A).
- Cost supports are $\emptyset$ and $\{(1, 1)\}$, which differ (B).
- For $a = b$: optimality plus feasibility force $u_0 + v_0 = 0$, $u_1 + v_1 = a$, so every optimal dual is the gauge shift by $t = u_0$ of $(0, a), (0, 0)$: unique mod gauge. For $a < b$ the pair $u = (0, b), v = (a - b, 0)$ is also optimal and is not a gauge shift of $(0, a), (0, 0)$ (it would need $t = 0$ from u_0 , but v_0 differs): not unique.

Pair 2 (nonzero): $C_{1,1}, C_{1,2}$: value $1/2$, same cost support, but different optimal-plan supports and dual structure (`main_nonzero`); this covers readers who object to the zero cost.

3 Lean

File `lean/Conjecture6680/Basic.lean` (about 250 lines, one marked statement block). The statement block defines everything above and `Claim := ConjectureA ∧ ConjectureB`; the theorems are `C6680.main : Claim` and `C6680.main_nonzero`. Axioms: only `propext`, `Classical.choice`, `Quot.sound` (see `verification/axioms.txt` after the pull-request build). No sorry, no native_`decide`.

Reuse. The one-parameter description of 2×2 plans and the diagonal/anti-diagonal support idea are adapted from our own accepted solution of conjecture 00000006320 (lemmas `Omega2_form`, `Omega2_bounds`); GPL-3.0, same author. The transport objects and duality are new.

4 Conventions and limits

Marginals are the uniform $(\frac{1}{2}, \frac{1}{2})$; the dual structure is read as uniqueness modulo the gauge, and “different” as unique versus non-unique. Readings not covered: dual structure read as the set of complementary-slackness supports of dual optimizers, or as the set of dual-feasible polytope faces; these are not formalized.